

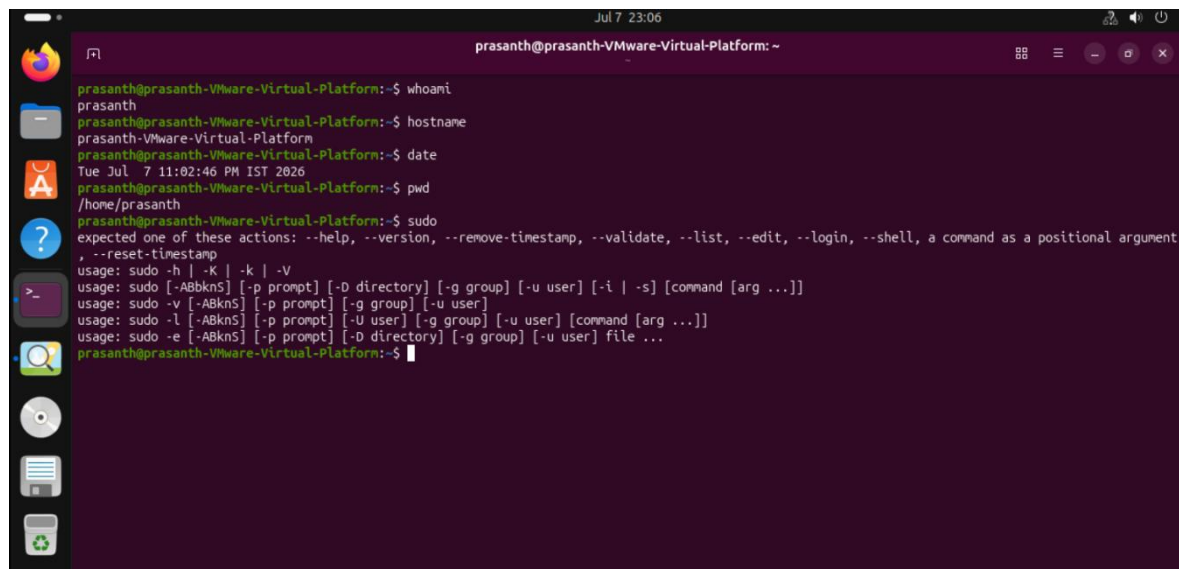
Part A: System Exploration

1. Open the terminal and find:

- Your current username
- Your hostname
- The current date and time
- The current working directory
- How to switch to root user
- What does sudo do

Commands :-

- whoami
- hostname
- pwd
- date
- uname -a
- sudo



```
prasanth@prasanth-VMware-Virtual-Platform:~$ whoami
prasanth
prasanth@prasanth-VMware-Virtual-Platform:~$ hostname
prasanth-VMware-Virtual-Platform
prasanth@prasanth-VMware-Virtual-Platform:~$ date
Tue Jul 7 11:02:46 PM IST 2026
prasanth@prasanth-VMware-Virtual-Platform:~$ pwd
/home/prasanth
prasanth@prasanth-VMware-Virtual-Platform:~$ sudo
expected one of these actions: --help, --version, --remove-timestamp, --validate, --list, --edit, --login, --shell, a command as a positional argument
, --reset-timestamp
usage: sudo -h | -K | -k | -V
usage: sudo [-ABbkns] [-p prompt] [-D directory] [-g group] [-u user] [-i | -s] [command [arg ...]]
usage: sudo -v [-ABkns] [-p prompt] [-g group] [-u user]
usage: sudo -l [-ABkns] [-p prompt] [-U user] [-g group] [-u user] [command [arg ...]]
usage: sudo -e [-ABkns] [-p prompt] [-D directory] [-g group] [-u user] file ...
prasanth@prasanth-VMware-Virtual-Platform:~$
```

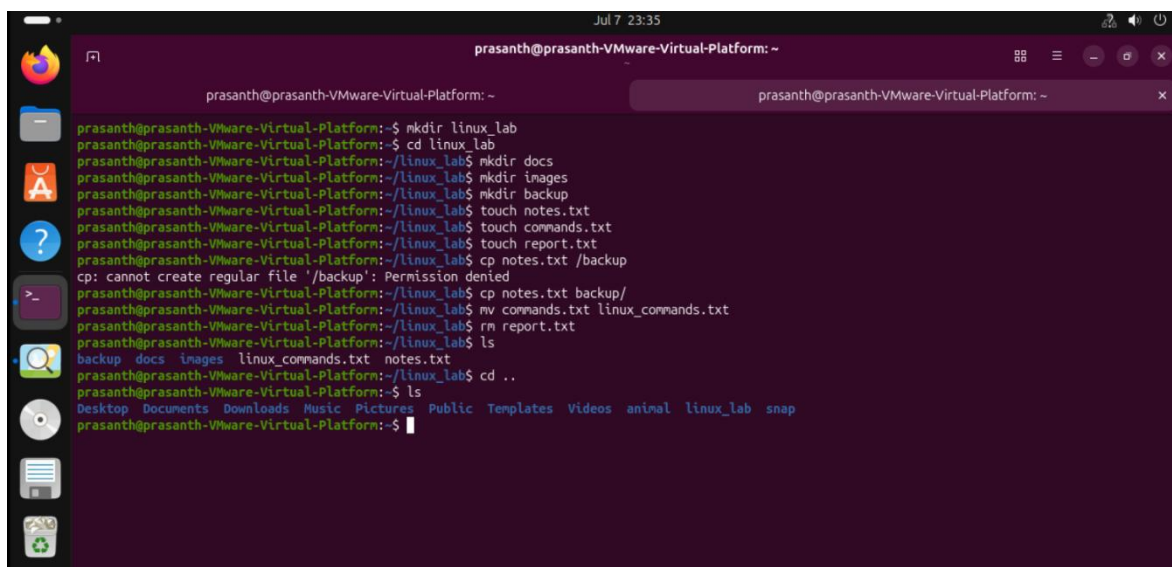
Part B: File and Directory Operations

Perform the following tasks:

1. Create a directory named linux_lab.
2. Inside it, create three directories:
 - o docs
 - o images
 - o backup
3. Create the following files:
 - o notes.txt
 - o commands.txt
 - o report.txt
4. Copy notes.txt to the backup directory.
5. Rename commands.txt to linux_commands.txt.
6. Delete report.txt

Commands :-

- mkdir
- touch
- cp
- mv
- rm



```
prasanth@prasanth-VMware-Virtual-Platform: ~  
prasanth@prasanth-VMware-Virtual-Platform: ~  
prasanth@prasanth-VMware-Virtual-Platform: ~$ mkdir linux_lab  
prasanth@prasanth-VMware-Virtual-Platform: ~$ cd linux_lab  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ mkdir docs  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ mkdir images  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ mkdir backup  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ touch notes.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ touch commands.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ touch report.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ cp notes.txt /backup  
cp: cannot create regular file '/backup': Permission denied  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ cp notes.txt backup/  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ mv commands.txt linux_commands.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ rm report.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ ls  
backup  docs  images  linux_commands.txt  notes.txt  
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab$ cd ..  
prasanth@prasanth-VMware-Virtual-Platform: ~$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  animal  linux_lab  snap  
prasanth@prasanth-VMware-Virtual-Platform: ~$
```

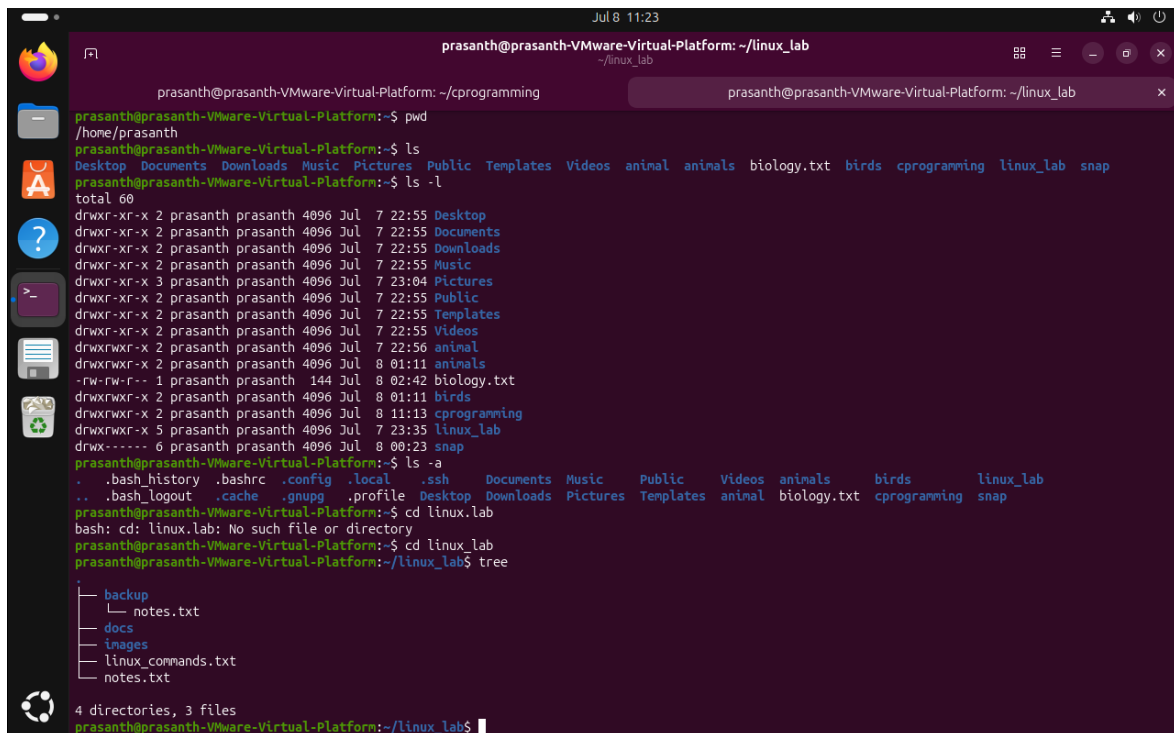
Part C: Navigation Commands

Practice:

- pwd
- ls
- ls -l
- ls -a
- cd
- tree

Tasks:

- Navigate between directories.
- Display hidden files.
- List files with detailed permissions.



```
prasanth@prasanth-VMware-Virtual-Platform: ~/linux_lab
prasanth@prasanth-VMware-Virtual-Platform: ~/programming

prasanth@prasanth-VMware-Virtual-Platform:~$ pwd
/home/prasanth
prasanth@prasanth-VMware-Virtual-Platform:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos animal animals biology.txt birds cprogramming linux_lab snap
prasanth@prasanth-VMware-Virtual-Platform:~$ ls -l
total 60
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Desktop
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Documents
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Downloads
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Music
drwxr-xr-x 3 prasanth prasanth 4096 Jul 7 23:04 Pictures
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Public
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Templates
drwxr-xr-x 2 prasanth prasanth 4096 Jul 7 22:55 Videos
drwxrwxr-x 2 prasanth prasanth 4096 Jul 7 22:56 animal
drwxrwxr-x 2 prasanth prasanth 4096 Jul 8 01:11 animals
-rw-rw-r-- 1 prasanth prasanth 144 Jul 8 02:42 biology.txt
drwxrwxr-x 2 prasanth prasanth 4096 Jul 8 01:11 birds
drwxrwxr-x 2 prasanth prasanth 4096 Jul 8 11:13 cprogramming
drwxrwxr-x 5 prasanth prasanth 4096 Jul 7 23:35 linux_lab
drwx----- 6 prasanth prasanth 4096 Jul 8 00:23 snap
prasanth@prasanth-VMware-Virtual-Platform:~$ ls -a
. .bash_history .bashrc .config .local .ssh Documents Music Public Videos animals birds linux_lab
.. .bash_logout .cache .gnupg .profile Desktop Downloads Pictures Templates animal biology.txt cprogramming snap
prasanth@prasanth-VMware-Virtual-Platform:~$ cd linux_lab
bash: cd: linux_lab: No such file or directory
prasanth@prasanth-VMware-Virtual-Platform:~$ cd linux_lab
prasanth@prasanth-VMware-Virtual-Platform:~/linux_lab$ tree
.
├── backup
│   └── notes.txt
├── docs
├── images
├── linux_commands.txt
└── notes.txt

4 directories, 3 files
prasanth@prasanth-VMware-Virtual-Platform:~/linux_lab$
```

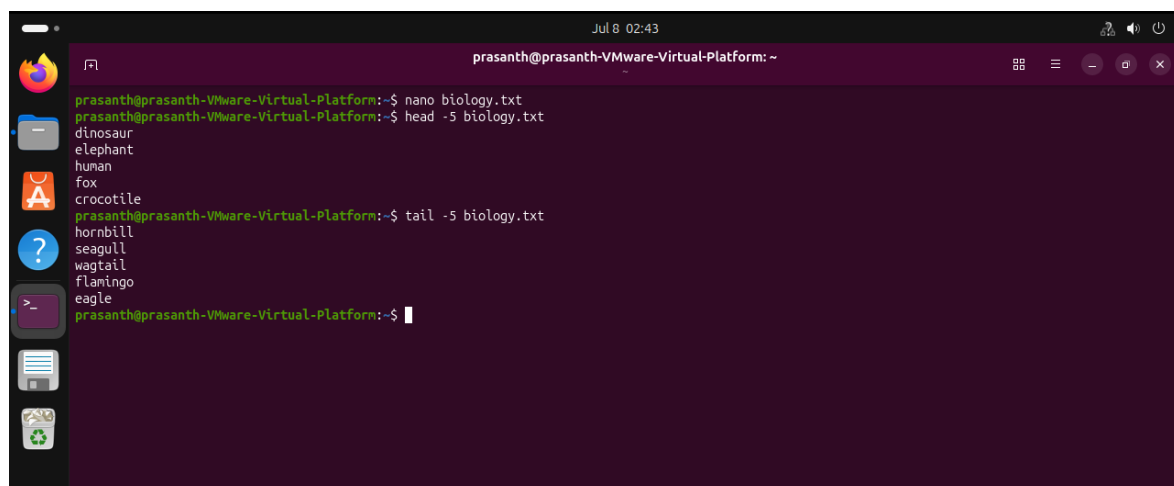
Part D: File Viewing Commands

Create a file with at least 20 lines and perform:

- cat
- more
- less
- head
- tail

Questions:

1. Show the first 5 lines.
2. Show the last 3 lines.
3. Explain the difference between cat, more, and less.



```
prasanth@prasanth-VMware-Virtual-Platform: ~  
prasanth@prasanth-VMware-Virtual-Platform:~$ nano biology.txt  
prasanth@prasanth-VMware-Virtual-Platform:~$ head -5 biology.txt  
dinosaur  
elephant  
human  
fox  
crocodile  
prasanth@prasanth-VMware-Virtual-Platform:~$ tail -5 biology.txt  
hornbill  
seagull  
wagtail  
flamingo  
eagle  
prasanth@prasanth-VMware-Virtual-Platform:~$
```

Cat:

Dumps the entire file at once to the terminal — no pausing, no scrolling control. Best for short files or piping output into other commands. Impractical for large files since everything flies past instantly

More:

Displays the file one screen (page) at a time. Press Space to move forward a page, enter to move one line. It only scrolls forward — in older/basic versions you can't go back up once you've passed a section.

Less:

It is also a pager, but more powerful: lets you scroll both forward and backward (arrow keys, Page Up/Down), search inside the file (/pattern), and doesn't need to load the whole file into memory — so it stays fast even on huge files. It's considered the modern replacement for more.

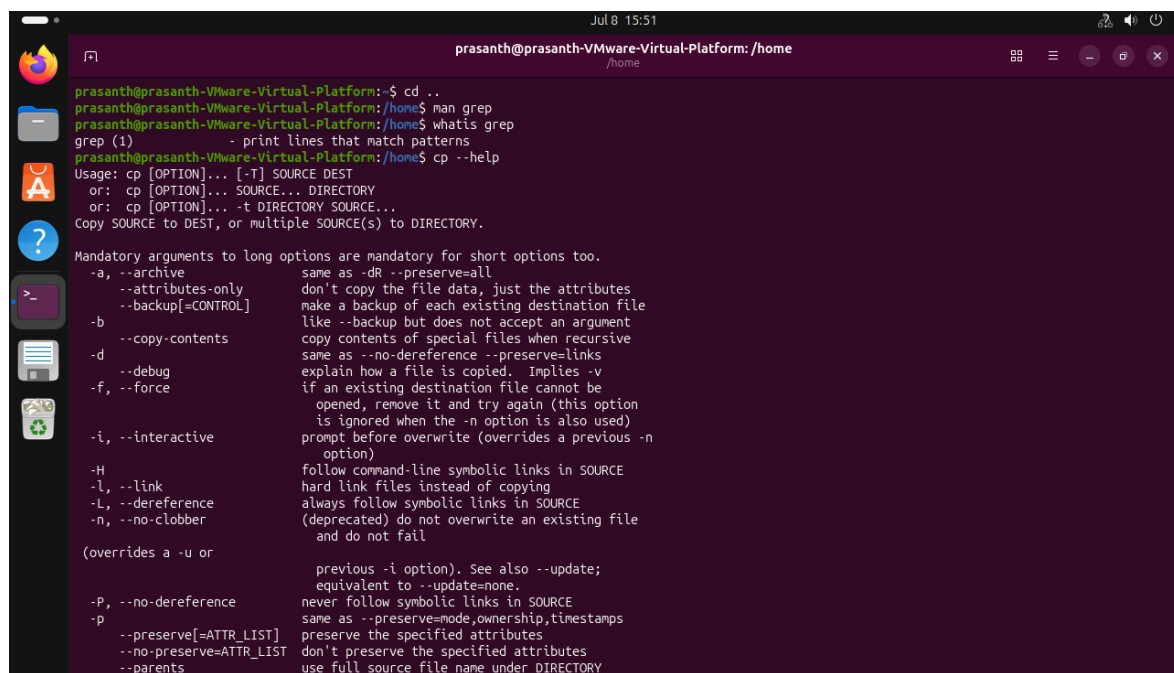
Part E: Help Commands

Use:

- man
- whatis
- which
- --help

Tasks:

1. Find the purpose of the grep command.
2. Find where the ls command is stored.
3. List three options available for the cp command.



```
prasanth@prasanth-VMware-Virtual-Platform:~$ cd ..
prasanth@prasanth-VMware-Virtual-Platform:/home$ man grep
prasanth@prasanth-VMware-Virtual-Platform:/home$ whatis grep
grep (1)          - print lines that match patterns
prasanth@prasanth-VMware-Virtual-Platform:/home$ cp --help
Usage: cp [OPTION]... [-T] SOURCE DEST
or: cp [OPTION]... SOURCE... DIRECTORY
or: cp [OPTION]... -t DIRECTORY SOURCE...
Copy SOURCE to DEST, or multiple SOURCE(s) to DIRECTORY.

Mandatory arguments to long options are mandatory for short options too.
-a, --archive                same as -dR --preserve=all
--attributes-only            don't copy the file data, just the attributes
--backup[=CONTROL]          make a backup of each existing destination file
-b                           like --backup but does not accept an argument
--copy-contents              copy contents of special files when recursive
-d                           same as --no-dereference --preserve-links
--debug                     explain how a file is copied. Implies -v
-f, --force                  if an existing destination file cannot be
                             opened, remove it and try again (this option
                             is ignored when the -n option is also used)
-i, --interactive            prompt before overwrite (overrides a previous -n
                             option)
-H                           follow command-line symbolic links in SOURCE
-l, --link                   hard link files instead of copying
-L, --dereference             always follow symbolic links in SOURCE
-n, --no-clobber              (deprecated) do not overwrite an existing file
                             and do not fail
                             (overrides a -u or
                             previous -i option). See also --update;
                             equivalent to --update=none.
-P, --no-dereference         never follow symbolic links in SOURCE
-p                           same as --preserve=mode,ownership,timestamps
--preserve[=ATTR_LIST]      preserve the specified attributes
--no-preserve=ATTR_LIST     don't preserve the specified attributes
--parents                    use full source file name under DIRECTORY
```

```
Jul 8 15:52
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/home

--parents          use full source file name under DIRECTORY
-R, -r, --recursive copy directories recursively
--reflink[=WHEN]   control clone/COW copies. See below
--remove-destination remove each existing destination file before
                   attempting to open it (contrast with --force)
--sparse=WHEN      control creation of sparse files. See below
--strip-trailing-slashes remove any trailing slashes from each SOURCE
                   argument
-s, --symbolic-link make symbolic links instead of copying
-S, --suffix=SUFFIX override the usual backup suffix
-t, --target-directory=DIRECTORY copy all SOURCE arguments into DIRECTORY
-T, --no-target-directory treat DEST as a normal file
--update[=UPDATE]   control which existing files are updated;
                   UPDATE={all,none,none-fail,older(default)}
                   equivalent to --update[=older]. See below
-u, --verbose       explain what is being done
-v, --keep-directory-symlink follow existing symlinks to directories
-x, --one-file-system stay on this file system
-Z                set SELinux security context of destination
                   file to default type
--context[=CTX]    like -Z, or if CTX is specified then set the
                   SELinux or SMACK security context to CTX
--help            display this help and exit
--version         output version information and exit

ATTR_LIST is a comma-separated list of attributes. Attributes are 'mode' for
permissions (including any ACL and xattr permissions), 'ownership' for user
and group, 'timestamps' for file timestamps, 'links' for hard links, 'context'
for security context, 'xattr' for extended attributes, and 'all' for all
attributes.

By default, sparse SOURCE files are detected by a crude heuristic and the
corresponding DEST file is made sparse as well. That is the behavior
selected by --sparse=auto. Specify --sparse=always to create a sparse DEST
file whenever the SOURCE file contains a long enough sequence of zero bytes.
Use --sparse=never to inhibit creation of sparse files.

UPDATE controls which existing files in the destination are replaced.
'all' is the default operation when an --update option is not specified,
and results in all existing files in the destination being replaced.
```

```
Jul 8 15:52
prasanth@prasanth-VMware-Virtual-Platform: /home
/home

corresponding DEST file is made sparse as well. That is the behavior
selected by --sparse=auto. Specify --sparse=always to create a sparse DEST
file whenever the SOURCE file contains a long enough sequence of zero bytes.
Use --sparse=never to inhibit creation of sparse files.

UPDATE controls which existing files in the destination are replaced.
'all' is the default operation when an --update option is not specified,
and results in all existing files in the destination being replaced.
'none' is like the --no-clobber option, in that no files in the
destination are replaced, and skipped files do not induce a failure.
'none-fail' also ensures no files are replaced in the destination,
but any skipped files are diagnosed and induce a failure.
'older' is the default operation when --update is specified, and results
in files being replaced if they're older than the corresponding source file.

When --reflink[=always] is specified, perform a lightweight copy, where the
data blocks are copied only when modified. If this is not possible the copy
fails, or if --reflink=auto is specified, fall back to a standard copy.
Use --reflink=never to ensure a standard copy is performed.

The backup suffix is '~', unless set with --suffix or SIMPLE_BACKUP_SUFFIX.
The version control method may be selected via the --backup option or through
the VERSION_CONTROL environment variable. Here are the values:

    none, off      never make backups (even if --backup is given)
    numbered, t    make numbered backups
    existing, nil  numbered if numbered backups exist, simple otherwise
    simple, never  always make simple backups

As a special case, cp makes a backup of SOURCE when the force and backup
options are given and SOURCE and DEST are the same name for an existing,
regular file.

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/cp>
or available locally via: info '(coreutils) cp invocation'
prasanth@prasanth-VMware-Virtual-Platform:/home$ which ls
/usr/bin/ls
prasanth@prasanth-VMware-Virtual-Platform:/home$ cp
```